

Shehab Ahmed

Hamilton, ON • +1 (905) 920-8894 • Shehab.f.ahmed@gmail.com • in/shehab-ahmed-automation • github.com/shehabmac

SUMMARY OF QUALIFICATIONS

- Industrial automation expertise: PLC programming, HMI design and implementation, motion control, and robotic process automation
- Hands-on experience integrating microcontrollers and IoT communication protocols for industrial applications
- Skilled in real-time networking protocols for system communication and remote equipment monitoring
- SCADA systems, digital electronics, and process control instrumentation experience
- Strong troubleshooting and problem-solving skills, with a focus on process optimization and automation efficiency
- Strong communicator; skilled in cross-functional collaboration and project coordination

EDUCATION

McMaster University — Bachelor of Technology (Co-op), Faculty of Engineering

Sep 2021 – Dec 2026

- Integrated 4.5-year Automation Engineering Technology program; GPA: 9.0/12
- Business Certificate and Advanced Chemical Engineering Diploma, Mohawk College

EXPERIENCE

RELEVANT EXPERIENCE

Linamar Corporation – Skyjack Inc. – Automation Engineer Co-op

Jan – Aug 2024 & Apr – Jul 2025

- Engineered robotic welding processes and RAPID code in ABB RobotStudio for automated production
- Leveraged AI-assisted tooling to standardize and rename 1,000+ air and weld targets in RAPID code to company standards
- Optimized Tube Laser, Micro Levels/Bases automation for Skyjack scissor and boom lifts via automated GMAW production
- Designed and implemented FactoryTalk View Studio HMIs integrated with PLCs, streamlining manufacturing control systems
- Optimized PLC-HMI communication and remote monitoring using Ethernet/IP and TCP/IP, reducing downtime by 10%
- Enhanced system reliability by resolving electrical and instrumentation issues via schematics and technical documentation
- Improved device performance by 23% through preventive maintenance and testing, ensuring compliance with safety protocols

McMaster University – C++ Procedural and Object-Oriented Programming Teaching Assistant

Jan – Dec 2023

- Guided 100+ students in C++ programming projects, offering debugging strategies and object-oriented design principles
- Led 20 lab sessions, providing structured tutorials and coding exercises that improved lab report accuracy and code efficiency
- Assessed 700+ lab reports, identifying coding errors and optimization opportunities while upholding academic integrity

ADDITIONAL EXPERIENCE

American International School in Egypt (AISE) – Teaching Assistant

Sep 2025 – Apr 2026

- Delivered an American curriculum for French/ICT via lesson preparation, maintaining a safe, structured learning environment
- Provided one-on-one and group academic support, promoting positive behavior through respectful classroom management

BookTips – Digital Marketing and Sales Intern

Jul – Aug 2022

- Increased sales by 2% by segmenting audiences and personalizing email campaigns, improving customer engagement
- Executed 50+ weekly cold calls using Audiense and Twitter/X marketing, increasing engagement by 6%

TECHNICAL SKILLS

- **Automation and Control Systems:** PLC programming (RSLogix, RSLinx, Studio5000, Fanuc Ladder 3), HMI (LabVIEW, FactoryTalk View), 6-axis robotic and motion control programming (ABB, Fanuc), P&IDs and instrumentation calibration
- **IIoT and Predictive Maintenance:** Device/network architecture design (MQTT, edge/cloud computing), vibration analysis and FFT-based condition monitoring, Siemens IDE and Ignition SCADA dashboards
- **Software Development and Machine Learning:** Python, C++, SQL, HTML, CSS, JavaScript, NumPy, OpenCV, neural networks
- **CAD and Simulation Tools:** AutoCAD, SolidWorks, MATLAB, Simulink
- **Manufacturing and Process Engineering:** Chemical process design (mass and energy balance), Lean Six Sigma
- **Office Tools:** MS Office suite (Excel, Macros, Word, SharePoint, Teams)

PROJECTS

SAMQ – 6-Axis Robotic Arm (Capstone Project)

- Architecting the control software for a 6-axis robotic arm as part of a 3-person capstone team in C++ and JSON with platform.io
- Implementing motion routines to enable the arm to grip objects via varying currents from point A to point B

FPGA Elevator

- Devised and programmed a 3-level elevator system using FPGA and Verilog, implementing real-time control logic
- Integrated P/B, BCD decoder, 7-segment display, motor driver, and LEDs for floor selection, movement, and status indication

IoT Smart Desk – Cloud-Integrated Automation System

- Developed an Arduino-powered smart desk with automated height adjustment, fan control, and ultrasonic user detection
- Engineered a stepper motor system and cloud-integrated LabVIEW HMI for user customization, enhancing ergonomics

Automated Pet Feeder – IoT-Enabled Smart Feeding System

- Developed a pet feeder with RFID-based identification, Wi-Fi-enabled MQTT communication, and automated food dispensing
- Calibrated ultrasonic sensors to monitor food levels and servo motors for portion control, enabling remote monitoring

EXTRA-CURRICULAR ACTIVITIES

Bachelor of Technology Association – Automation Representative and VP Administration

Jan 2023 – Dec 2024

McMaster University – Management Factor Competition Finalist

Feb 2022